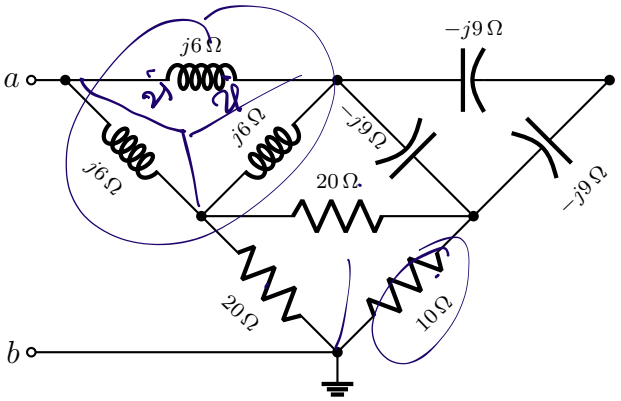
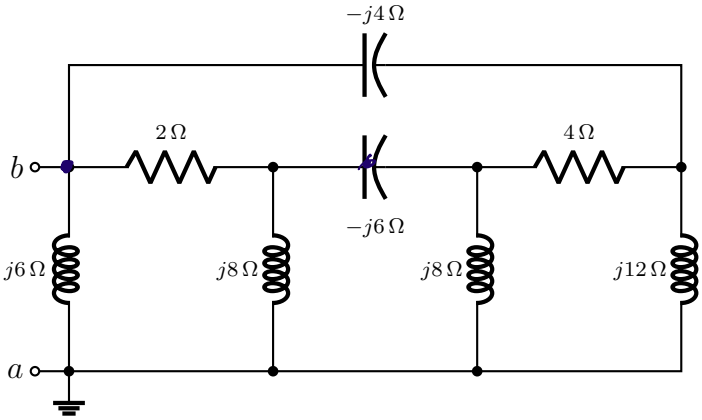
EE2015: Electric Circuits and Networks

Tutorial 4

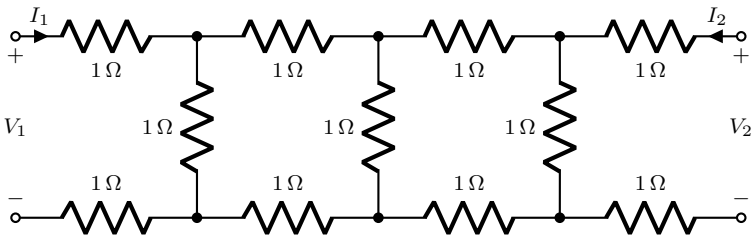
(September 12, 2025)

1. Calculate the impedance $\mathbf{Z}_{ab}$ of the one-port networks shown below.



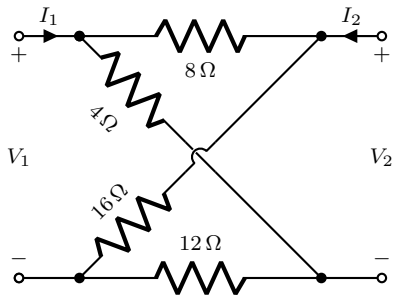


2. For the two-port network shown on the right find the impedance and admittance parameters.



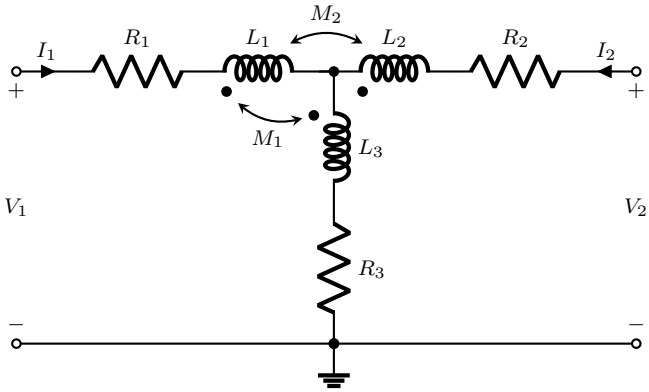
Determine the z- and y-parameters of the two-port network shown on the left.

3.

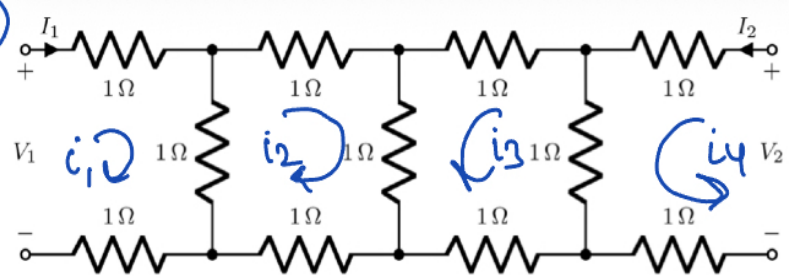


4.

Find the entries of $\mathbf{Z}(s)$ of the two-port network shown on the right.



(2)



$$V_1 - I_1 - I_1 + I_2 - I_1 = 0$$

$$I_2 - I_1 + I_2 + I_2 + I_3 + I_2 = 0$$

$$I_3 + I_2 + I_3 + I_3 + I_3 - I_4 = 0$$

$$V_2 - I_4 - I_4 + I_3 - I_4 = 0$$

$$3I_1 - I_2 = V_1 \qquad 4I_2 + I_3 - I_1 = 0$$

$$3I_4 - I_3 = V_2 \qquad 4I_3 + I_2 - I_4 = 0$$

$$4\{3I_1 - V_1\} + \{3I_4 - V_2\} - I_1 = 0$$

$$4\{3I_4 - V_2\} + \{3I_1 - V_1\} - I_4 = 0$$

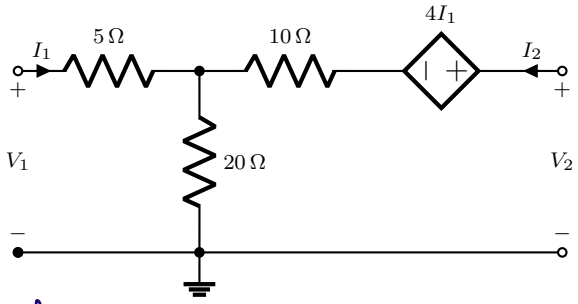
$$\Rightarrow 11I_1 + 3I_4 = 4V_1 + V_2$$

$$\Rightarrow 3I_1 + 11I_4 = V_1 + 4V_2$$

$$\Rightarrow I_1 = \frac{41V_1 - V_2}{112} \qquad I_2 = \frac{41V_2 - V_1}{112}$$

$$\Rightarrow Y = \begin{bmatrix} \frac{41}{112} & \frac{-1}{112} \\ \frac{-1}{112} & \frac{41}{112} \end{bmatrix} \qquad Z = Y^{-1} = \frac{112}{15}\begin{bmatrix} \frac{41}{112} & \frac{1}{112} \\ \frac{1}{112} & \frac{41}{112} \end{bmatrix} = \begin{bmatrix} \frac{41}{15} & \frac{1}{15} \\ \frac{1}{15} & \frac{41}{15} \end{bmatrix}$$

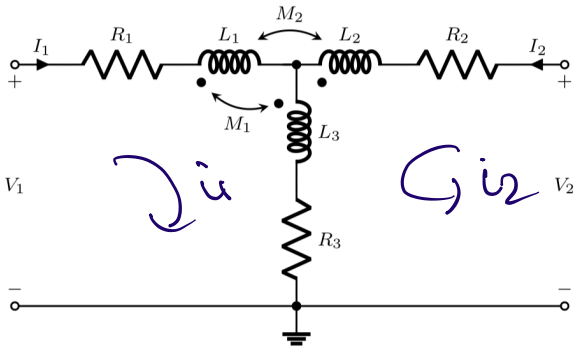
5. Find the z-parameters for the two-port network shown on the right.



$V_1 = 5I_1 + 20(I_1 + I_2) = 25I_1 + 20I_2$

$V_2 = 4I_1 + 10I_2 + 20(I_1 + I_2) = 24I_1 + 30I_2$

$$z = \begin{bmatrix} 25 & 20 \\ 24 & 30 \end{bmatrix}$$

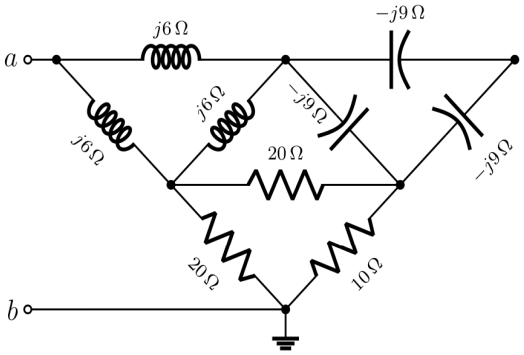


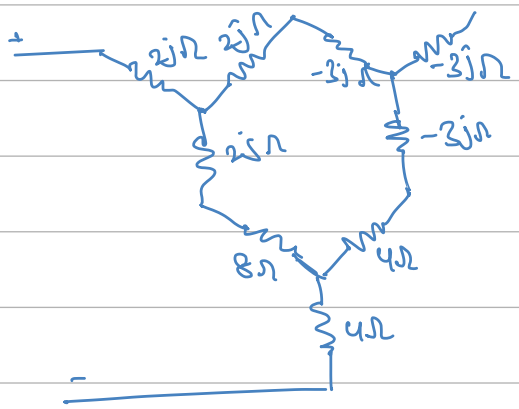
$V_1 = I_1R_1 + sL_1I_1 + sM_1(I_1 + I_2) + sM_1I_1 + sL_3(I_1 + I_2) + (I_1 + I_2)R_3 - sM_2I_2$

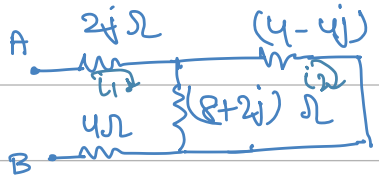
$V_2 = I_2R_2 + sL_2I_2 - sM_2I_1 + sL_3(I_1 + I_2) + sM_1I_1 + (I_1 + I_2)R_3$

$V_1 = I_1[R_1 + sL_1 + 2sM_1 + sL_3 + R_3] + I_2[sM_1 + sL_3 - sM_2 + R_3]$

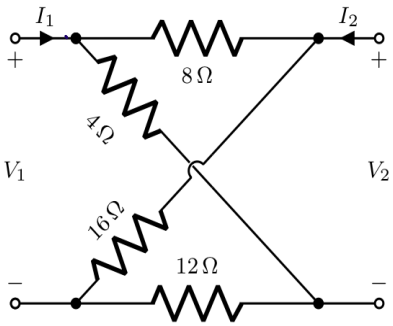
$V_2 = I_1[sM_2 + sL_3 + sM_1 + R_3) + I_2[R_2 + sL_2 + sL_3 + R_3]$

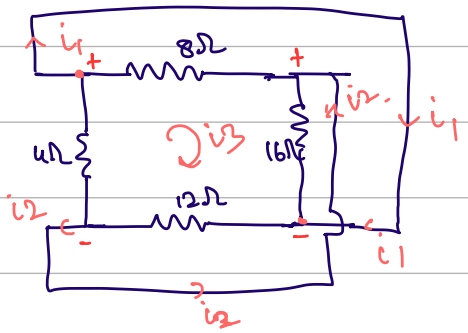






$$\frac{(4-4j)(8+2j)}{12-2j}+4+2j=\frac{20-12j}{6-j}+4+2j=\frac{(46-4j)(6+j)}{(6-j)(6+j)}=\frac{280+22j}{37}$$





$$-V_1=(i_1-i_3-i_2)16+(i_1-i_3)8$$

$$V_2=16(i_3+i_2-i_1)+12(i_3+i_2)$$

$$12(i_3+i_2)+4i_3+8(i_1+i_3)+16(i_2+i_3-i_1)=0$$

$$\Rightarrow 40i_3+28i_2-24i_1=0 \Rightarrow i_3=\frac{3i_1}{5}-\frac{7}{10}i_2$$

$$-V_1=24(i_1-i_3)-16i_2=24\left(\frac{2i_1}{5}+\frac{7}{10}i_2\right)-16i_2$$

$$=9.6i_1-0.8i_2$$

$$V_2=28\left(\frac{3i_1}{5}+\frac{3}{10}i_2\right)-16i_1=-0.8i_1+8.4i_2$$

$$\Rightarrow Z=\begin{bmatrix}9.6 & -0.8\\ -0.8 & 8.4\end{bmatrix}$$

$$Y=Z^{-1}$$